Comptia N+ Assignment

[N+ Network Fundamental and building networks]

Introduction to Network Devices

Overview of switches, routers, hubs, firewalls, and access points.Practical (1 hour):Identify ports and functions of network devices in a setup.

Task: Create a table comparing network devices and their features.

❖ **1. Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.**

➢ **Aim**

- To study the main functions and typical ports of switches, routers, hubs, firewalls, and wireless access points used in a home Wi-Fi network.

➢ **Introduction**

- Network devices are hardware components that connect computers, mobile phones, printers, and other devices to a network. Each device performs a specific function, such as forwarding data, connecting different networks, providing wireless connectivity, or protecting the network from unauthorized access.
- In a typical home network, a router connects the home network to the Internet, a switch connects multiple wired devices, an access point provides Wi-Fi connectivity, and a firewall filters network traffic. A hub

is an older device that connects multiple wired devices by broadcasting incoming data to all its ports.

- **Comparison Table of Network Devices**

Network Device	Main Functions	Typical Ports / Interfaces	Common Use in a Home Wi-Fi Setup
Switch	Connects multiple wired devices within a LAN and forwards data to the appropriate port using MAC addresses.	Multiple Ethernet ports, commonly RJ-45 (e.g., 5-port or 8-port switches).	Connects desktop computers, printers, smart TVs, and other wired devices.
Router	Connects different networks, forwards IP packets, and commonly provides NAT, DHCP, and routing functions.	WAN/Internet Ethernet port, LAN Ethernet ports, and sometimes USB.	Connects the home network to the Internet and distributes network connectivity.
Hub	Connects multiple devices in a LAN and sends incoming data to all other ports. It does not intelligently forward frames using MAC addresses.	Multiple Ethernet ports, commonly RJ-45.	An older way to connect wired devices; largely replaced by switches.

Firewall	Monitors and filters incoming and outgoing network traffic according to security rules.	Physical interfaces may include WAN, LAN Ethernet, and management ports; software firewalls may have no dedicated physical ports.	Protects the home network from unwanted or unauthorized network traffic.
Wireless Access Point (AP)	Provides Wi-Fi connectivity and bridges wireless devices to the wired LAN.	Usually an Ethernet RJ-45 port; some models support PoE (Power over Ethernet).	Connects phones, laptops, tablets, and other Wi-Fi devices to the home network.

- **Explanation of Common Network Ports**
  1. **RJ-45 Ethernet Port:** Used to connect network devices through Ethernet cables.
  2. **WAN Port:** Connects a router to an Internet modem or an upstream network.
  3. **LAN Port:** Connects local wired devices, such as computers and printers, to the home network.
  4. **USB Port:** Available on some routers for supported storage devices or other functions.
  5. **Management Port:** Available on some professional network devices for configuration and administration.

➢ **Result**

- The main functions and typical ports of switches, routers, hubs, firewalls, and wireless access points were studied and compared in a table.

❖ **2. Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.**

➢ **Aim**

- To identify three network devices and label their important ports, including Ethernet, console, power, and antenna ports wherever available.

➢ **Introduction**

- Network devices are hardware components used to connect computers and other devices in a network. Different devices have different ports for data communication, power supply, configuration, and wireless connectivity. Identifying these ports helps in installing, configuring, and troubleshooting network devices.

➢ **Identification of Network Devices and Ports**

➢ **A. Router**

➢ A router connects different networks and forwards data packets between them. A home Wi-Fi router commonly contains the following ports and components:

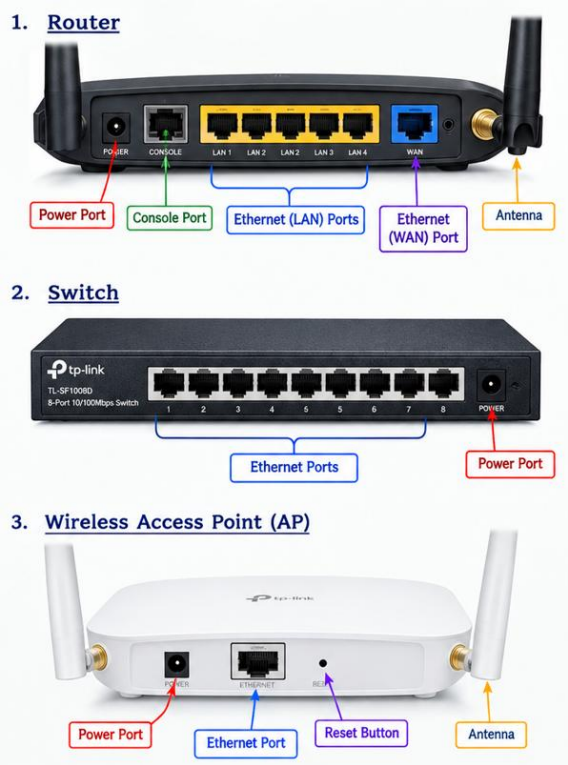
1. **WAN Port**: Connects the router to the modem or Internet service provider.
2. **LAN (Ethernet) Ports:** Connect computers, switches, and other wired devices.
3. **Power Port:** Connects the router to its power adapter.
4. **Console Port:** Available on some routers for configuration and management.
5. **Antennas:** Provide wireless Wi-Fi signals in routers that have external antennas.

- **B. Switch**
- A switch connects multiple wired devices within a Local Area Network (LAN). It forwards data to the appropriate port using MAC addresses.
  1. **Ethernet Ports:** Connect computers, printers, routers, and other network devices.
  2. **Power Port:** Supplies electrical power to the switch.
  3. **Console Port:** Available on some managed switches for configuration and troubleshooting.

- **C. Wireless Access Point (AP)**
- A wireless access point allows Wi-Fi-enabled devices to connect to a wired network.
  1. **Ethernet Port:** Connects the access point to a router or network switch.
  2. **Power Port:** Connects the access point to a power adapter.
  3. **PoE Support:** Some access points receive power through their Ethernet cable using Power over Ethernet (PoE).

4. **Antennas:** Transmit and receive wireless signals. Some access points have internal antennas.

- **Diagram**
- The labelled diagram shows a router, a switch, and a wireless access point with their important ports and components.



- **Result**
- Three network devices—router, switch, and wireless access point—were identified, and their main ports and functions were studied.

❖ **3. Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.**

➢ When I use Instagram on my phone, different network devices help me access its services. Instagram is a popular social media application used to share photos, videos, and messages over the Internet.

1. **Wireless Access Point:** It connects my phone to the home Wi-Fi network and allows data to travel wirelessly.
2. **Router:** It forwards data between my home network and the Internet, allowing my phone to communicate with Instagram's servers.
3. **Firewall:** It filters network traffic according to security rules and helps protect the network from unauthorized connections.

❖ **4. Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help. Think about device capacity, coverage, and traffic handling.**

➢ **Problem**

- The college Wi-Fi network is slow and sometimes disconnects. This may happen because too many students are connected to the same wireless access point, the Wi-Fi coverage is weak, or the network devices cannot handle the traffic efficiently.

- **Possible Reason**
  - One possible reason is that the existing wireless access points do not have enough capacity to handle the large number of students and their devices. When many users access the Internet simultaneously, the access point may become overloaded, causing slow speeds and frequent disconnections. Poor Wi-Fi coverage may also contribute to the problem.

- **Suggested Solution: Upgrade the Wireless Access Points**
- The college can upgrade its existing access points to modern, high-capacity Wi-Fi access points.
  1. **Device Capacity:** Modern access points can support more connected devices and handle multiple users more efficiently.
  2. **Coverage:** Properly placed access points can provide better Wi-Fi coverage and reduce weak-signal areas.
  3. **Traffic Handling:** Modern access points can manage wireless traffic more efficiently, improving connectivity when many students use the network simultaneously.
- Switches and routers should also be checked. A switch with insufficient capacity or a router that cannot handle the total network traffic may become a bottleneck. If either device is responsible, upgrading that device may be more effective.

- **Result**
- Upgrading the wireless access points, if they are the cause of the problem, can improve Wi-Fi coverage, support more users, and reduce slow connections and disconnections.